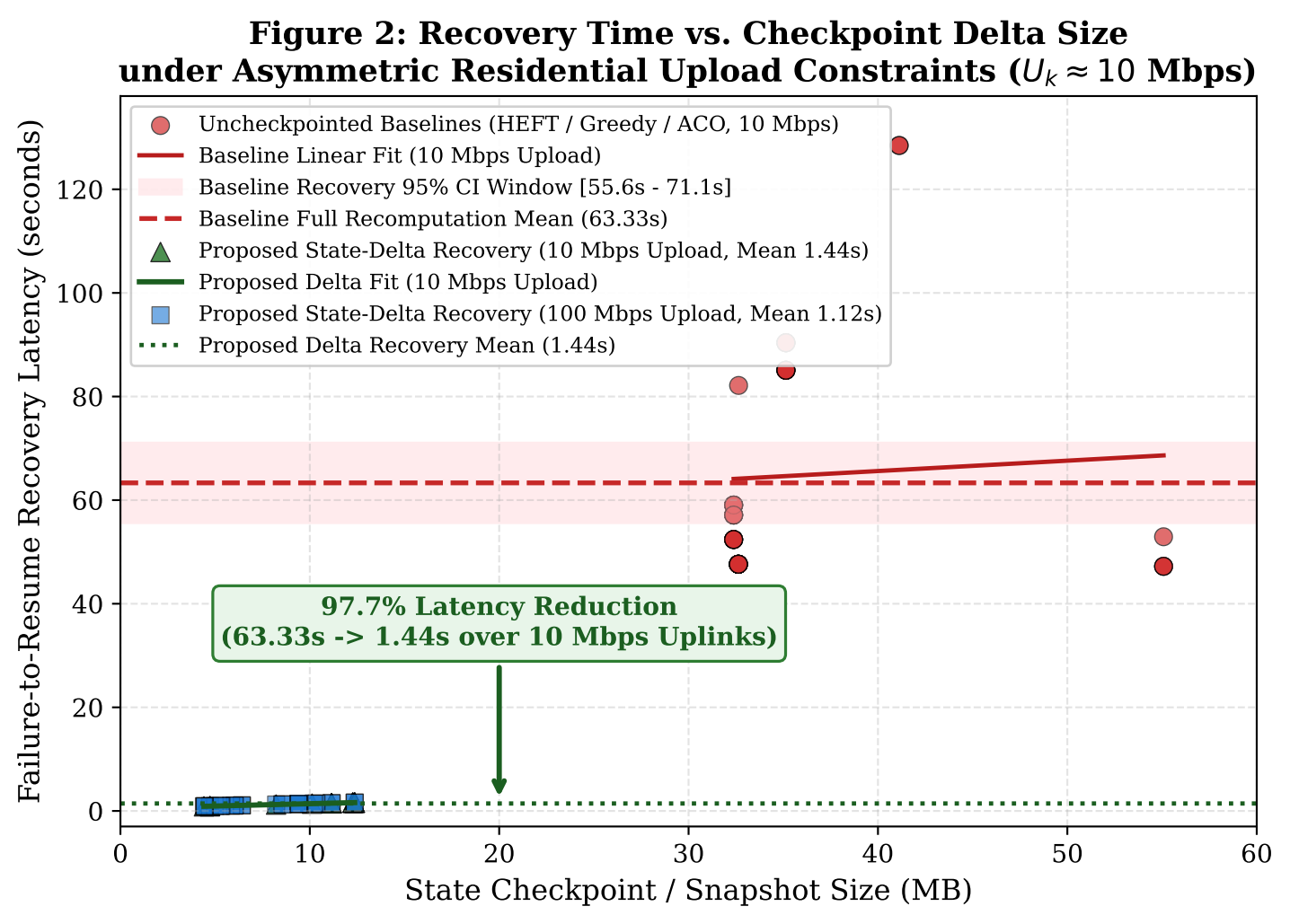
Figure 2: Recovery Time vs. Checkpoint Delta Size
under Asymmetric Residential Upload Constraints ($U_k \approx 10$ Mbps)
Uncheckpointed Baselines (HEFT / Greedy / ACO, 10 Mbps)
Baseline Linear Fit (10 Mbps Upload)
Baseline Recovery 95% CI Window [55.6s - 71.1s]
Baseline Full Recomputation Mean (63.33s)
Proposed State-Delta Recovery (10 Mbps Upload, Mean 1.44s)
Proposed Delta Fit (10 Mbps Upload)
Proposed State-Delta Recovery (100 Mbps Upload, Mean 1.12s)
Proposed Delta Recovery Mean (1.44s)
97.7% Latency Reduction
(63.33s -> 1.44s over 10 Mbps Uplinks)
Failure-to-Resume Recovery Latency (seconds)
State Checkpoint / Snapshot Size (MB)
0
20
40
60
80
100
120
0
10
20
30
40
50
60